Submission CALC1-PS1-SUB01

Question CALC1-PS1-Q01
scratch: show step
1. $f'(x)=2x(6x-1)+(x^2+3)6$
2. $f'(3)=6(17)+(12)6=24$
=> FINAL: 24 (check units/interpretation)

Question CALC1-PS1-Q02
scratch: show step
1. $\int 3x\,dx = (3/2)x^2$
2. $[3/2\, x^2]_0^2 = 6$
=> FINAL: 6 (check units/interpretation)

Question CALC1-PS1-Q03
scratch: show step
1. $2w+2l=22 \Rightarrow l=11-w$
2. $A(w)=w(11-w)$; at $w=8$, $A=24$
3. A maximum requires $A'(w)=0$, so the stated dimensions are not assumed optimal without that check.
=> FINAL: $A(w)=w(11-w)$; $A(8)=24$; check $A'(w)=0$ (check units/interpretation)

Question CALC1-PS1-Q04
scratch: show step
1. $x^2-16=(x-4)(x+4)$
2. cancel $x-a$ only after identifying $x\neq a$
3. $\lim_{x\to 4} x+4=8$
=> FINAL: 8 (check units/interpretation)

Question CALC1-PS1-Q05
scratch: show step
1. Critical point: $f'(c)=0$ or $f'(c)$ is undefined (when f is defined).
2. A sign change in f' from + to – indicates a local maximum.
=> FINAL: critical point: $f'(c)=0$ or undefined; + to – sign change gives a local maximum (check units/interpretation)

Question CALC1-PS1-Q06
scratch: show step
1. $f'(x)=2x(4x-1)+(x^2+4)4$
2. $f'(3)=6(11)+(13)4=28$
=> FINAL: 28 (check units/interpretation)